Auction-Price Prediction for Used Tractors

Benchmarks and Generalisation

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1 Abstract

This MSc dissertation asks whether a nonlinear model improves used-tractor auction-price estimates beyond a transparent regression benchmark. The work constructs a sale sample from repeated listings, handles sparse specifications and models prices across five currencies without pooling incompatible monetary units. The modelling contribution is an evaluated data and regression pipeline, rather than a new learning algorithm.

Cleaning reduces 354,607 listings to 22,857 distinct sale records. A currency- and price-stratified random split allocates 15,999 records to training, 3,429 to validation and 3,429 to testing. CatBoost is selected by validation error on a currency-scaled log-price target. It and the quadratic comparator are refitted on training plus validation before test evaluation.

Test median absolute percentage error (MedAPE) is 31.96% for CatBoost and 40.18% for quadratic regression: an 8.22-percentage-point difference, or a 20.45% relative reduction in MedAPE. A paired 10,000-resample observation bootstrap gives a 95% interval of [6.60, 10.06] percentage points for the difference. Only 47.6% of CatBoost predictions fall within 30% of the recorded price. MedAPE is 22.55% for common model series and 44.02% for rare or unseen series; GBP error is 57.52%.

Both archived fitted models were restored and reproduce all 3,429 saved predictions each. The bootstrap intervals also reproduce. Raw-listing cleaning and the full hyperparameter search were not rerun. Transaction dates and physical-asset identifiers are unavailable; identical measured predictor profiles cross the split. The research decision is to retain CatBoost as the stronger cross-sectional benchmark, while requiring time- and asset-disjoint evaluation before a future-market valuation claim.

2 Introduction

Used agricultural equipment is heterogeneous. Two tractors of the same manufacture year can differ in usage, model series, equipment and market. A price estimate must therefore be assessed against observed sales and a credible simpler model. A visually plausible prediction or a high fitted association is insufficient evidence of useful valuation.

The first problem is defining the target sample. Listing feeds include repeated observations and records without a realised auction sale. Keeping all of them would overstate sample size and could teach a model to predict asking prices rather than realised prices. The second problem is limited specification coverage: missing horsepower, transmission and usage information constrain what any model can explain.

This study compares additive and quadratic regression, a decision tree, XGBoost and CatBoost. The quadratic model is the principal final-test comparator because it adds nonlinear age and usage terms while remaining a transparent regression specification. CatBoost can additionally learn interactions and categorical structure.

The evaluation separates model selection from the reported final comparison. It also asks where average performance hides unreliable predictions. Currency and training-support slices reveal errors that a pooled headline conceals. The study supports a within-sample cross-sectional result; the missing transaction timeline prevents an evaluation of price drift or genuinely future sales.

3 Data Construction and Missing Information

3.1 From listing feed to sale records

The documented cleaning funnel is shown below. These counts are recovered from the executed analysis and audit tables, not from a fresh raw-feed cleaning run.

Table 1: Documented sample construction. Distinct records do not establish distinct physical assets.

Stage	Records
Original listings	354,607
Sold auction records	40,326
After manufacture-year validity check	40,317
After substantive-record deduplication	22,857

Nine records with manufacture year above 2026 are excluded at the validity stage. Deduplication removes 17,460 repeated records, using substantive record characteristics and price rather than scrape identifiers. A unique physical-asset identifier is unavailable. The resulting rows are best described as distinct recorded sales, not independently verified unique machines.

3.2 Coverage and assumptions

Usage hours are missing in 25.1% of the cleaned sample; 74.9% describes availability. Horsepower is missing in 87.1% and transmission in 84.0%, so those fields are investigated descriptively but excluded from the final model. Usage is retained with median imputation and an explicit missingness flag.

The final inputs include manufacture year, hours, country, make, model series, currency, drive and equipment flags. Missing drive is assigned four-wheel drive under the supplied domain guidance. This is an assumption, not an observed specification. Equipment flags describe recorded availability; missing or inconsistently recorded details can limit their interpretation.

4 Target, Features and Evaluation Boundary

4.1 Currency-scaled log prices

For positive sale price p_i in currency c, let $\ell_i = \log p_i$. The initial training sample supplies a currency-specific median m_c and interquartile range s_c. The model target is

$$z_i = \frac{\ell_i - m_c}{s_c}, \qquad \widehat{p}_i = \exp(m_c + s_c \widehat{z}_i). \tag{1}$$

The transformation permits a common regression target without pretending that one dollar, pound and euro are interchangeable. It is not an exchange-rate adjustment or purchasing-power conversion. The inverse transformation gives a point prediction; it includes no additional lognormal smearing correction.

4.2 Training-only preprocessing

The original training partition defines category support. Makes and countries appearing fewer than 30 times and model series appearing fewer than five times are grouped as OTHER. The same rules are applied to validation and test records. Currency scaling also remains fixed at the initial training estimates. The final refit uses all 19,428 development rows; numerical median imputation is recalculated on that development sample.

The 70/15/15 split uses seed 42 and strata formed from currency and within-currency log-price quintiles. Stratification balances the observed price distribution across samples. It is a retrospective split design, not an operational rule requiring access to a future record's price. All candidate selection is based on validation performance.

4.3 What the split can establish

There is no auction transaction date. The available observation timestamp describes collection, not the underlying sale date, and cannot substitute for it. A random holdout therefore measures performance among rows from the observed sample. No temporal generalisation result is available. The October audit also finds 135 of 3,429 test rows with an identical measured predictor profile in the initial training sample. Different machines can share those measurements, so this is not proof of duplicated physical assets; it prevents a strong claim of asset-disjoint validation.

5 Models and Selection

5.1 Transparent controls and nonlinear challengers

Additive regression provides a simple reference. Quadratic regression adds degree-two terms for the numerical year and hours predictors, alongside one-hot categorical features and equipment flags. A decision tree and two gradient-boosting families test whether richer nonlinear effects and interactions improve validation accuracy.

CatBoost uses categorical information directly. The selected configuration has 700 boosting iterations, learning rate 0.03, depth seven, L2 leaf penalty five, absolute-error loss and seed 42. The restored model uses a single prediction thread for verification. The project evaluates an established algorithm; the contribution lies in sample construction, controls and error analysis.

Table 2: Saved validation selection scores. Lower scaled MAE is better.

Model	Validation scaled MAE
CatBoost	0.35230
XGBoost	0.37109
Decision tree	0.39425
Quadratic regression	0.40790
Additive regression	0.41170

Selection uses mean absolute error in the scaled log-price target, not final native-price MedAPE. These objectives differ. After selection, CatBoost and the quadratic comparator are refitted on the same training-plus-validation records. Their untouched-at-fitting test predictions supply the final comparison. The test sample has since been examined, including the present diagnostic review, so subsequent changes need an independent confirmation sample.

6 Overall Results and Uncertainty

For each positive price, define $a_i = 100|\widehat{p}_i - p_i|/p_i$. MedAPE is median$(a_i)$. It describes the typical relative mistake, not mean percentage error, explained variance, business savings or a confidence interval for an individual valuation.

Table 3: Final comparison on 3,429 held-out records. Intervals condition on the fitted models and observed sample.

Measure	Estimate	Bootstrap 95% interval
CatBoost MedAPE	31.96%	[30.56, 33.38]%
Quadratic MedAPE	40.18%	[38.76, 42.20]%
Improvement, percentage points	8.22	[6.60, 10.06]
Relative reduction in MedAPE	20.45%	[16.71, 24.23]%

A paired observation bootstrap uses 10,000 resamples with seed 42. Each resample selects the same record indices for both models, retaining the paired comparison. It does not refit the models, repeat selection, account for future market drift or establish independence of related machine profiles. A clustered or time-block analysis requires identifiers that are missing here.

CatBoost places 34.2% of predictions within 20% of the recorded price, and 47.6% within 30%. Consequently, more than half exceed a 30% relative error. An improvement over regression does not by itself make the system a sufficiently precise pricing tool. No deployment tolerance, asymmetric loss function or realised decision benefit has been validated.

7 Where Predictions Become Unreliable

7.1 Market and currency differences

Table 4: Currency-specific final-test errors. Native MAE uses the currency named in each row; amounts must not be added or averaged across currencies.

Currency	Rows	CatBoost MedAPE	Quadratic MedAPE	Native MAE
AUD	66	35.50%	34.36%	31,620
CAD	1,270	40.04%	50.43%	29,225
EUR	496	18.05%	24.92%	7,594
GBP	357	57.52%	71.47%	7,373
USD	1,240	26.10%	30.40%	12,793

CatBoost improves the pooled relative metric but does not win every currency comparison: the quadratic model has lower AUD MedAPE in a small sample. GBP has the weakest CatBoost relative performance. Currency can encode market composition as well as monetary scale; these slices do not isolate a causal country or currency effect.

7.2 Training support

Common model series account for 1,605 test rows and have CatBoost MedAPE of 22.55%. The 1,824 rare or unseen rows have MedAPE of 44.02%. Their within-30% rates are 60.4% and 36.3%, respectively. The grouping uses support in the original training partition, not test outcomes.

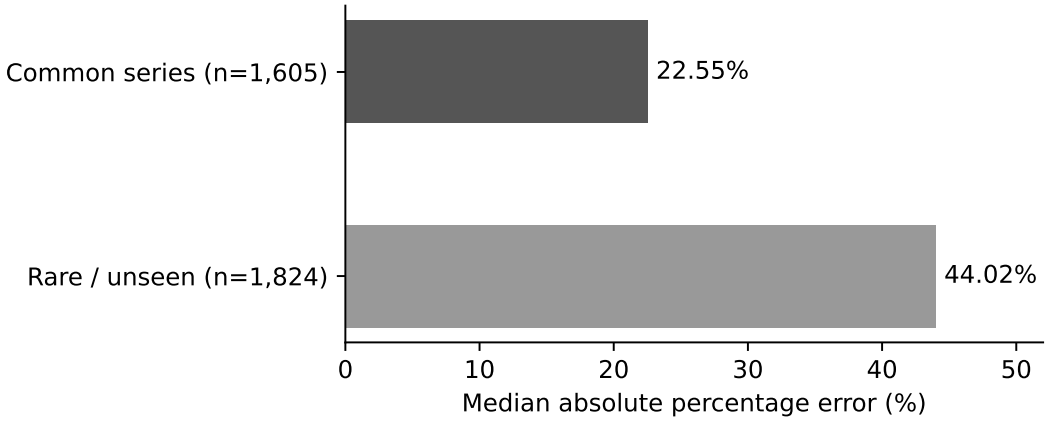

Figure 1: Median percentage error by model-series support in the initial training sample. These retrospective slices identify where the fitted model is less reliable; they are not independent model-selection experiments.

This support gap suggests a future abstention or human-review policy, but no confidence threshold or policy benefit is validated here. Permutation importance is likewise a measure of predictive dependence in this fitted system, not evidence that adding an equipment feature causes a price premium.

8 Verification and Reproducibility

The October 2026 audit restores the archived CatBoost and quadratic fitted models and rebuilds the saved split and training-only transformations. Each model reproduces all 3,429 saved price predictions, with maximum absolute numerical differences below 6×10^{-11} native units. All four published bootstrap intervals reproduce from 10,000 paired resamples. Nine explicit evidence checks pass.

Two archived cleaned-data copies agree exactly as parsed tables, although their file bytes differ. The source archive and cleaned-input SHA-256 fingerprints are recorded in the companion verification JSON. The restored prediction environment uses Python 3.12.14, NumPy 2.3.5, pandas 2.2.3, scikit-learn 1.8.0 and CatBoost 1.2.10. No model-load warnings were emitted in this environment.

The audit verifies fitted-model prediction replay and result arithmetic. It does not rerun raw-listing cleaning, the full candidate search or all model training. It provides no new time-disjoint or asset-disjoint evaluation. Historical result tables remain distinct from newly recomputed diagnostic values.

The public repository contains the original dissertation, aggregate results, this report and analysis notebooks. The revised training notebook contains source only; the original executed version remains in the pinned source snapshot. Original listing records and archived fitted-model files are not redistributed in the repository. A reader can inspect the analysis and aggregate evidence, but full retraining still requires access to the source dataset. Dependencies are installed from the requirements file before opening the notebooks; bare installation statements have been removed from Python analysis cells.

9 Conclusion and Next Experiment

The study demonstrates how to turn a repeated listing feed into an auditable regression problem, compare nonlinear models with simpler controls and assess failure modes beyond one overall score. CatBoost has a materially lower test MedAPE than quadratic regression on the observed cross-sectional holdout. The restored models and paired bootstrap reproduce the saved result.

The practical qualification is equally clear: typical errors remain large, most predictions are outside a 30% tolerance, rare series are substantially harder, and random records do not establish future-market performance. The model should remain a research valuation benchmark until an appropriate decision tolerance and independent evaluation are defined.

The next experiment should collect actual sale timestamps and stable asset or auction-lot identifiers, freeze a later market period and group related assets before splitting. Preserve a simple quadratic control, test make/model support and currency slices, and assess review or abstention rules only on a separate development period. Prediction intervals and asymmetric valuation losses should be evaluated against a stated user decision. This is proposed work, not a completed result.

10 Sources and Report Provenance

This report presents the author's used-tractor auction-price MSc project in a plain research-report format. Its assembly date does not establish the project's actual start or completion dates. Counts and results are reconciled to the dissertation, executed notebooks, archived fitted models and output tables.

References

[1] Hussain, F. H. Predicting Used-Tractor Auction Prices. Pinned source version, executed notebooks and original dissertation.

[2] Hussain, F. H. Research decision and reproducibility scope. Companion aggregate verification record. October 2026.